

Aerosol Optical Depth (AOD) variation over Ahmedabad

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Research Context

Aerosols influence global climate through direct and indirect radiative forcing mechanisms. This work addresses some of the AOD variation patterns. Aerosols cause **direct radiative forcing** by scattering (cooling) or absorbing (warming) sunlight, while **indirect forcing** happens as they change cloud properties (making them brighter/longer-lasting via cloud condensation nuclei), leading to more reflection and cooling. Overall present-day forcing is a net cooling effect counteracting some greenhouse gas warming.

Key Terms

- **Aerosols:** Suspended particles including dust, smoke, pollution, and sea salt in the atmosphere
- **Aerosol Optical Depth (AOD):** Columnar integration of extinction of sunlight by aerosols
- **Direct Irradiance:** Solar radiation reaching Earth's surface directly
- **Extraterrestrial Irradiance:** Solar radiation at the top of atmosphere before attenuation

Why This Research Matters: AOD data provides essential insights for air quality assessment, climate impact quantification, source apportionment between anthropogenic and natural aerosols, and evidence-based environmental policy development for rapidly industrializing cities.

Sun Photometer

Helps us get the **Direct Solar Irradiance** at specific wavelengths. Measures **Aerosol Optical Depth (AOD)**, also called Aerosol Optical Thickness (AOT), that quantifies how 'hazy' the atmosphere is by measuring how much sunlight is blocked by aerosols.



Working Principle: Selective light absorption

Measures sunlight reaching ground and compares it to how much sunlight *would* be at the top of atmosphere.

- **Tracking alignment:** Manually point the device exactly at the sun.
- **Baffle:** The sensor is placed at the bottom of narrow tube. This ensures that only parallel rays hit the sensor, blocking out scattered light.
- **Detector:** Photodiodes (light sensors) that convert light energy into an electrical voltage.

Spectral Wavelength Analysis



Sunlight is a mix of all colors - The device extracts data for specific wavelengths to **identify the type of aerosols**.

- **How it separates light:** Interference Filters placed in front of the photodiodes.
  - An interference filter blocks all light *except* a narrow band (10nm in our case).
- **Reason for Separation:** Different particles block different colors.
  - **Small particles** (smoke, smog) block blue/green light (short wavelengths) more effectively.
  - **Large particles** (dust, sand) block red/infrared light (long wavelengths) roughly equally.

Compare AOD at specific wavelengths: to determine if the haze is from a forest fire (small particles) or a dust storm (large particles).

How sun photometer evaluates AOD

According to **Bouguer-Lambert-Beer Law**, light's irradiance diminishes as it traverses through a medium. This concept is used for calculating Aerosol Optical Depth (AOD), which quantifies the total reduction of solar radiation caused by atmospheric particles.

$$I = I_0 \cdot \left(\frac{R_0}{R}\right)^2 \cdot e^{-m \cdot \tau}$$

$$V = k \cdot I$$

Where,

- I: Direct normal irradiance
- $I0$: **Extraterrestrial Irradiance**
- V: Raw signal measured by photometer
- $V0$ **Zero-Air-Mass Voltage** (the signal the device *would* read if it were in space)
- $R0/R$ Correction for the Earth-Sun distance (accounts for *~3.4%* seasonal variation in V0 due to Earth's elliptical orbit)
- m **Relative Air Mass** (accounts for relative change in path length as light traverses through the atmosphere (Noon m 1; Sunset m 1) >
- k: Constant which depends on wavelength

Total Optical Depth:

$$\tau = \frac{1}{m}\left[\ln(V_0) - \ln(V) + 2\ln\left(\frac{R_0}{R}\right)\right]$$

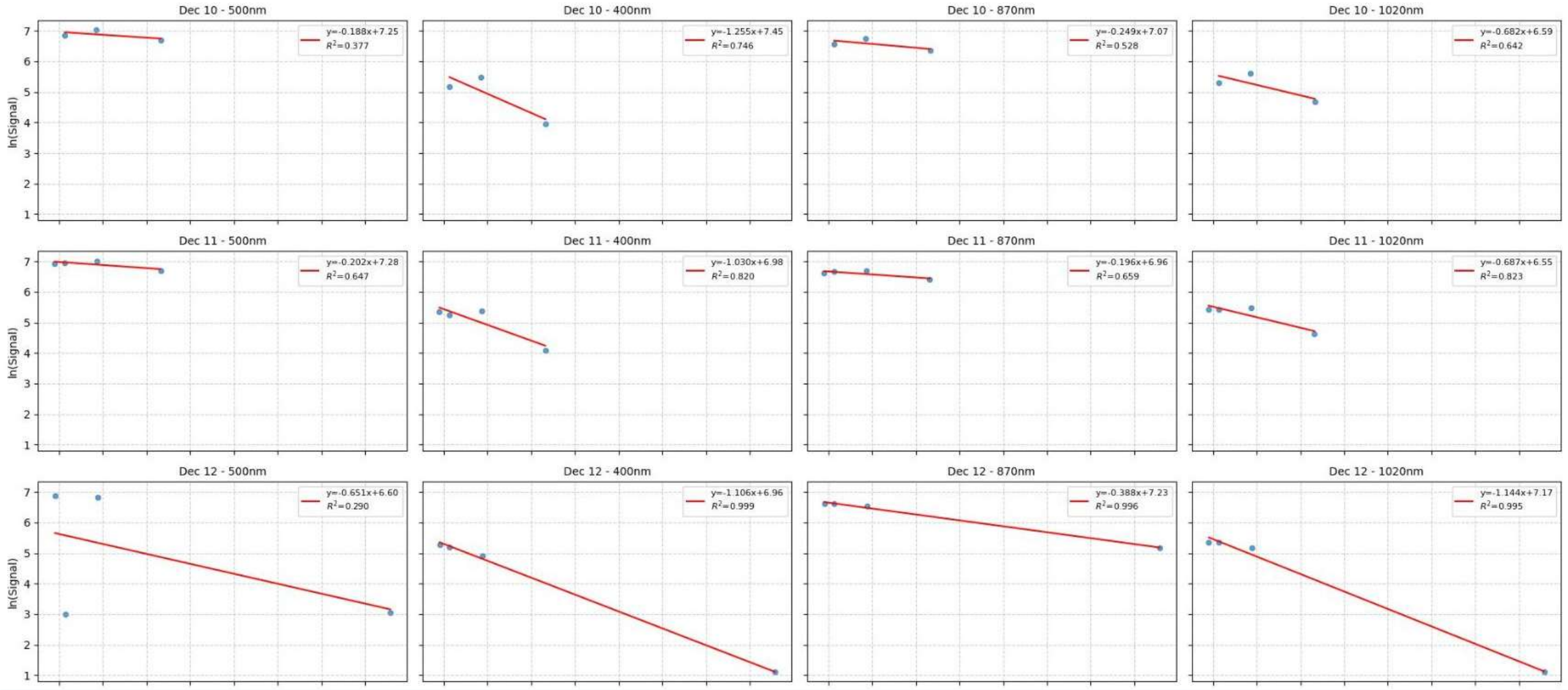
*Note: τ is the total optical depth that includes blocking by Air Molecules (Rayleigh scattering), Water vapor, Ozone, some gases (NO2, CO2, CH4), and Aerosols. These factors need to be subtracted to get the **aerosol optical depth**.*

But what about the signal Vo *to be read* in space?

Since it is impossible to physically place every field sun photometer in space to measure the sun's raw intensity, **Langley plot** allows us to extrapolate ground-based data to zero atmosphere.
So, Vo is the **calibration constant** derived from its intercept.

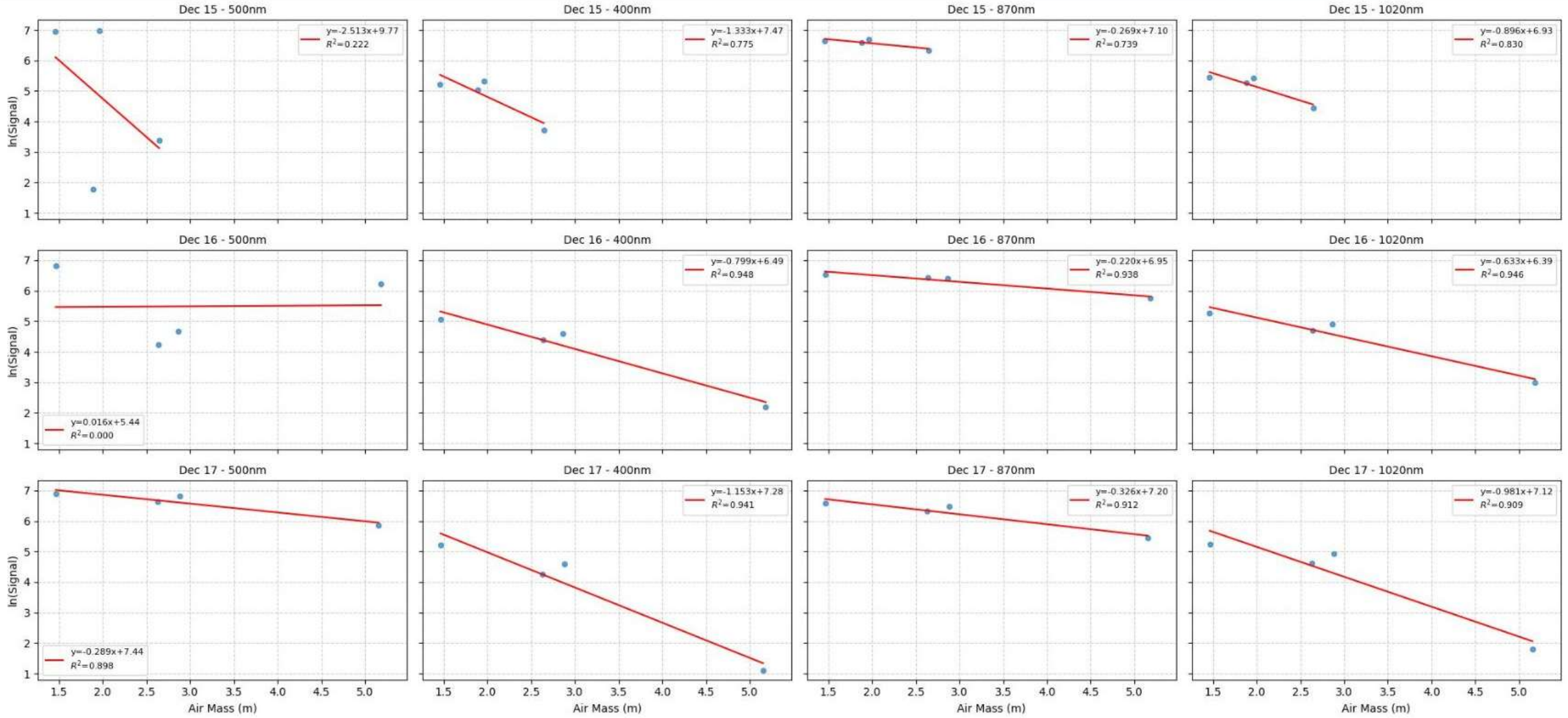
$$\ln(V) = -m \cdot \tau + [\ln(V_0) + 2\ln(\frac{R_0}{R})]$$

Langley Plots (for first 2 weeks) – ln(V) vs Relative Air Mass



To be continued

Langley Plots (for first 2 weeks) – ln(V) vs Relative Air Mass



Drawbacks

1. **Pointing Error:**
   - *Issue:* Because it's handheld, if the sensor is not perfectly aligned with sun, the measured voltage drops.
   - *Result:* Lower V makes the device think *more* light was blocked by the air, resulting in an **artificially high AOD**.
   - *Fix:* Always take multiple scans and use the **highest** signal reading, not the average.
2. **Cloud Contamination:**
   - *Issue:* Thin clouds drift in front of the sun and block light.
   - *Result:* Device interprets the cloud as aerosol pollution.
   - *Fix:* Watch the stability, usually aerosols change over hours where clouds change fast. Rapid jumps AOD: likely clouds, not pollution.
3. **Filter Degradation:**
   - *Issue:* Interference filters inside the lens degrade over time due to heat and UV exposure.
   - *Result:* Over months/years, The calibration constant V_0 changes, the readings may drift.
4. **Temperature Sensitivity:**
   - *Issue:* The photodiodes and electronics behave differently when hot.
   - *Result:* The "dark current" (noise) increases, potentially skewing readings if the device isn't temperature-compensated.

Further Research

Seasonal AOD pattern

Whether driven by coarse-mode mineral dust transported by wind patterns or fine-mode anthropogenic aerosols

Size Distribution Patterns

Determine aerosol size characteristics using Ångström exponent calculation

Meteorological Correlations

Investigate relationships between AOD and wind speed, direction, relative humidity, temprature, and rainfall patterns

Dual-Instrument Validation

Comparing/ Correlating results of sun photometer and microaethalometer data for reliable aerosol loading trends

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Thank you! now it's time to address your valuable questions...